

LFP Preprocessing

Chenglu 2025.7

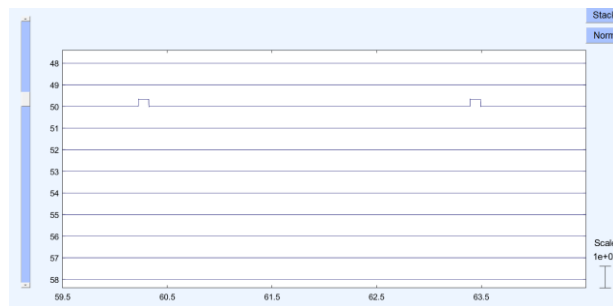
① Loading .edf file

② Add Event Marker:

1. Plot, 缩放调到 10000000 (7 个 0) 找到 marker channel (每次窗口只显示 10 个 channel, 时间 5s→这样比较好找)



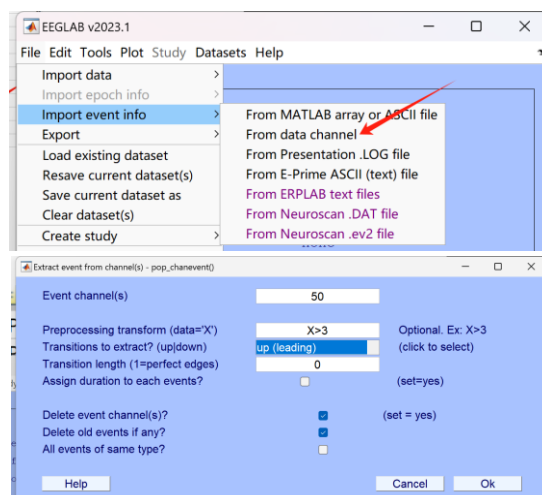
2. 找到这一个 channel 的数字序号: Plot - Setting- show number



(例如: 这里是 50)

#注意: 可能有多个 marker channel (这些 channel 信号是一样的), 这时候选取其中一个作为后续的 event 标记的 channel, 然后删去其它 marker channel

3. 加入 Event Marker



4. 运行 marker_clean.m

（由于从通道引入 marker 会存在一个信号被解析成多个 marker，因此通过这个脚本来去除一个 event 重复的 marker）（到这一步保存文件成_with_event.set）

③ Resampling:

Tool → Change sampling rate: 1000Hz

#注意：这里不用额外先低通滤波来防止混叠，因为 eeglab 内置了这一步骤：

Change the sampling rate

The most common use for Tools → Change sampling rate is to reduce the sampling rate to save memory and disk storage. A `pop_resample.m` window pops up, asking for the new sampling rate. The function uses MATLAB `resample.m` (in the Signal Processing toolbox- if you do not have this toolbox, it will use the slow MATLAB function `griddata`).



You do not need to change the sampling rate for processing the tutorial data since it is already at an acceptable sampling rate.

Before resampling, a low-pass filter at half the resampling frequency (so 64 Hz if you resample at 128 Hz) is applied to the data to avoid aliasing effects.

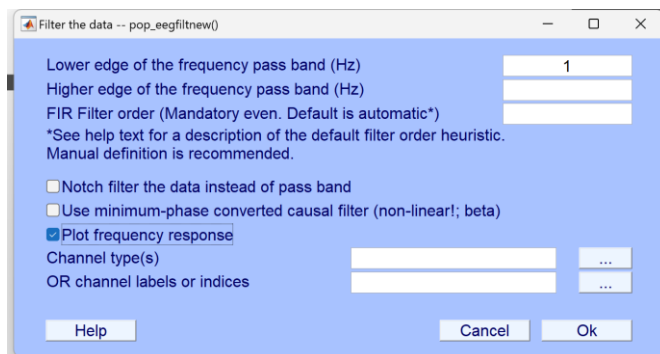
c. Resampling - EEGLAB Wiki

④ Filtering:

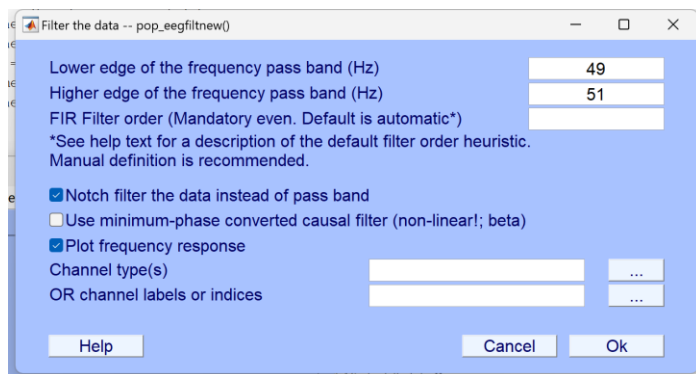
Tool → Filter the data: (FIR-1-200HZ)

1Hz 高通滤波跑一次

200Hz 低通滤波跑一次

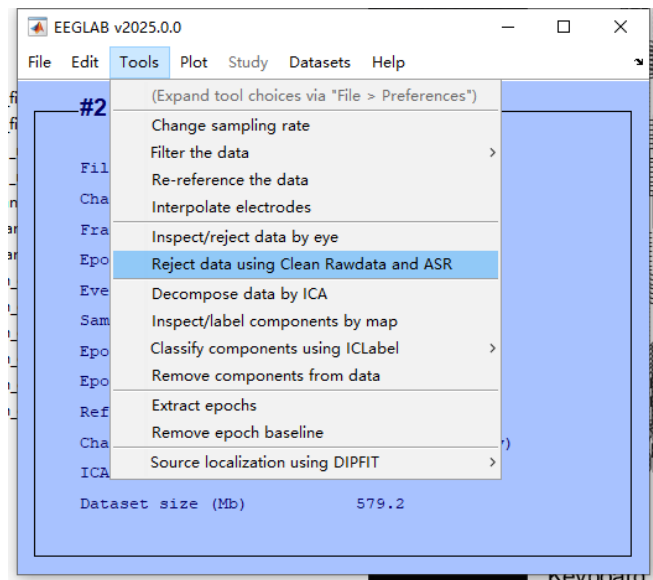


50Hz Notch

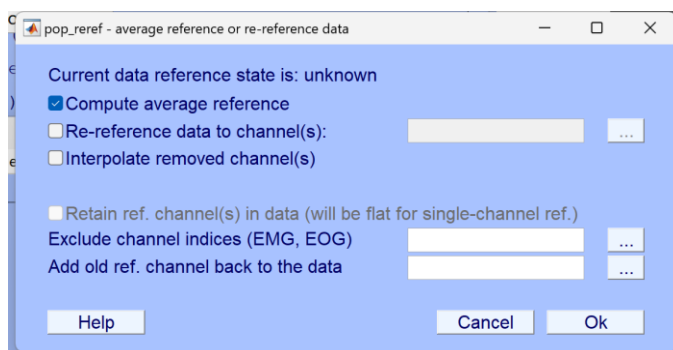


⑤ 删除坏导:

目前使用的是: Clean Raw Data (默认参数——平坦 5s, 超过 4 个标准差, 相关性 0.8)



⑥ Re-reference



⑦ Epoch

使用脚本 epoching.m

⑧ 删 trial

人工检查并删除

⑨ 时频分析

使用脚本 TFA.m